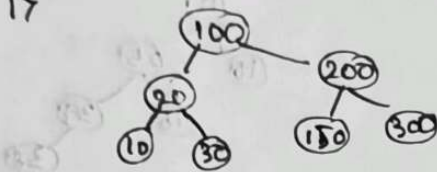


C03 - A73 Tree traversal challenge

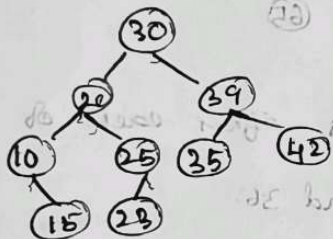
17



In order (L, N, R) = $\frac{10, 20, 30}{L} \quad \frac{100}{N} \quad \frac{150, 200, 300}{R}$

Preorder (N, L, R) = $\frac{100}{N} \quad \frac{10, 20, 30}{L} \quad \frac{150, 200, 300}{R}$

Post order (LRN) = $\frac{10, 20, 30}{L} \quad \frac{150, 300, 200}{R} \quad \frac{100}{N}$

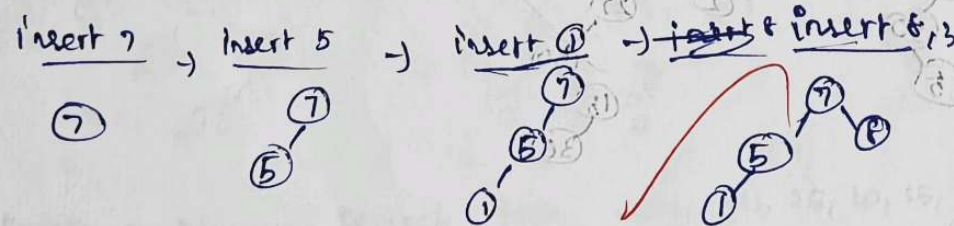


Inorder (L, N, R) = $\frac{10, 15, 20, 23, 25}{L} \quad \frac{30}{N} \quad \frac{35, 39, 42}{R}$

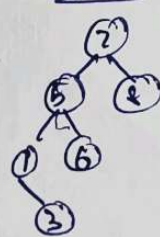
Preorder (N, L, R) = $\frac{30}{N} \quad \frac{10, 15, 20, 23, 25}{L} \quad \frac{35, 39, 42}{R}$

Postorder (LR, N) = $\frac{15, 20, 23, 25, 30}{L} \quad \frac{35, 42, 39}{R} \quad \frac{30}{N}$

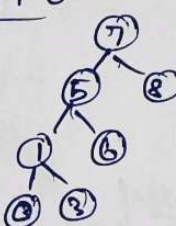
2) Binary search tree - 7, 5, 1, 8, 3, 6, 0, 9, 4, 2



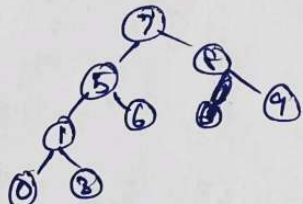
insert 6



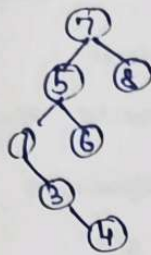
insert 0



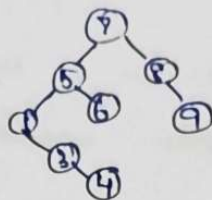
insert 9



insert 4



insert 2

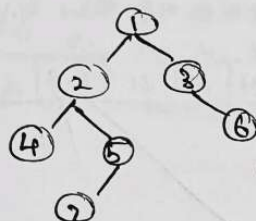


Inorder (L, N, R) :- $\frac{0, 1, 2, 3, 4, 5, 6}{L} \cdot \frac{7}{N} \cdot \frac{8, 9}{R}$

Preorder (N, L, R) :- $\frac{7}{N} \cdot \frac{5, 1, 2, 3, 4, 6}{L} \cdot \frac{8, 9}{R}$

Postorder (L, R, N) :- $\frac{0, 2, 4, 3, 1, 6, 5}{L} \cdot \frac{9, 8}{R} \cdot \frac{7}{N}$

57

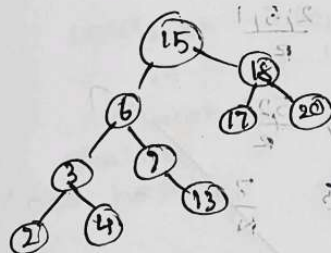


Inorder (L, N, R) = $\frac{4, 2, 7, 5}{L} \cdot \frac{1}{N} \cdot \frac{3, 6}{R}$

Preorder (N, L, R) = $\frac{1}{N} \cdot \frac{2, 4, 5, 7}{L} \cdot \frac{3, 6}{R}$

Postorder (L, R, N) = $\frac{4, 7, 5, 2}{L} \cdot \frac{6, 3}{R} \cdot \frac{1}{N}$

67

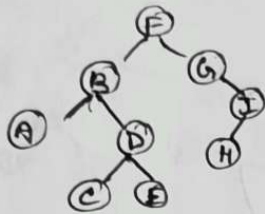


In order (L, N, R) = $\frac{2, 3, 4, 6, 7, 9, 13}{L} \cdot \frac{15}{N} \cdot \frac{17, 18, 20}{R}$

Pre order (N, L, R) = $\frac{15}{N} \cdot \frac{6, 3, 4, 7, 13, 9}{L} \cdot \frac{18, 17, 20}{R}$

Post order (L, R, N) = $\frac{2, 4, 3, 9, 13, 7, 6}{L} \cdot \frac{18, 20, 17}{R} \cdot \frac{15}{N}$

7



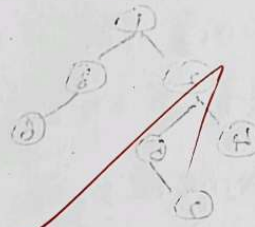
Inorder (L, N, R) = $\frac{A B C D E}{L} \frac{F}{N} \frac{G H I}{R}$

Preorder (N, L, R) = $\frac{F}{N} \frac{B A D C E}{L} \frac{G H I}{R}$

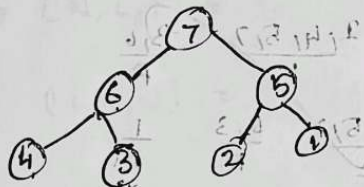
8 Inorder: $\frac{D B E F}{L} \frac{A}{N} \frac{G H C}{R}$

Pre order: $\frac{A B D E F}{N} \frac{G H C}{L} \frac{R}{R}$

Post order (L, R, N): $\frac{D F E B}{R} \frac{H G C}{R} \frac{A}{N}$



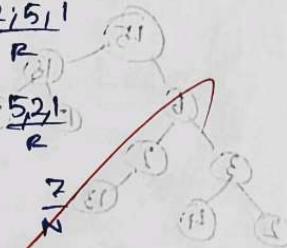
9



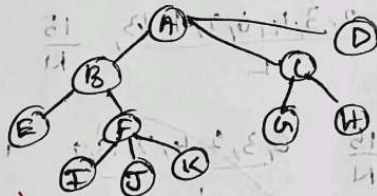
Inorder (L, N, R): $\frac{4, 6, 3}{L} \frac{7}{N} \frac{2, 5, 1}{R}$

Preorder (N, L, R): $\frac{7}{N} \frac{6, 4, 3}{L} \frac{5, 2, 1}{R}$

Postorder (L, R, N): $\frac{4, 3, 6}{L} \frac{2, 1, 5}{R} \frac{7}{N}$



10



Preorder (N, L, R): $\frac{A}{N} \frac{B, E, F, I, J, K}{L} \frac{C, G, H, D}{R}$

Post order (L, R, N): $\frac{E, I, J, K, F, B, G}{L} \frac{H, C, D}{R} \frac{A}{N}$

Level order: A, B, C, D, E, F, G, H, I, J, K